Сетевые технологии

Лабораторная работа №5

Просина Ксения Максимовна

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Раздел 1

1. Информация

1.1 Докладчик

- Просина Ксения Максимовна

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- Просина Ксения Максимовна
- Студент 3 курса

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1.1 Докладчик

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Раздел 2

2. Цель работы

2.1 Цель работы

Целью данной лабораторной работы является построение простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.

Раздел 3

3. Порядок выполнения работы

3.1 Моделирование простейшей сети на базе коммутатора в GNS3

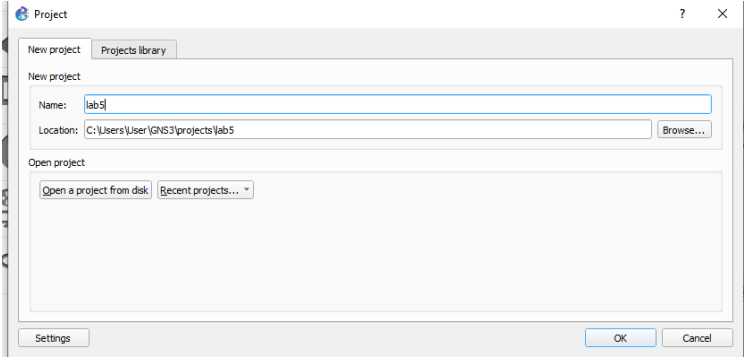


Рисунок 1: Запуск GNS3 и создание нового проекта

3.2 Моделирование простейшей сети на базе коммутатора в GNS3

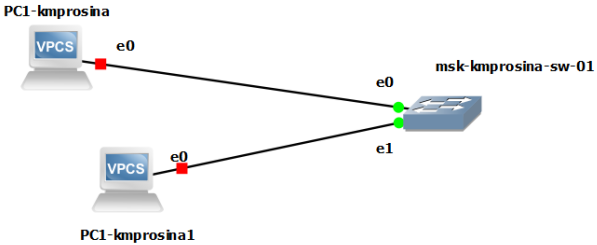


Рисунок 2: Добавление устройств в рабочую область GNS3

3.3 Моделирование простейшей сети на базе коммутатора в GNS3

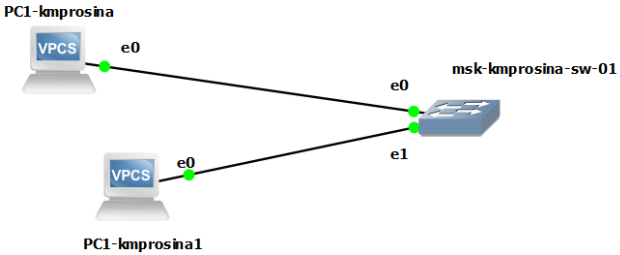


Рисунок 3: Настройка IP-адреса на PC1-user

3.4 Моделирование простейшей сети на базе коммутатора в GNS3

```
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
.  done
```

Рисунок 4: Сохранение конфигурации PC1-user

3.5 Моделирование простейшей сети на базе коммутатора в GNS3

```
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
.  done
```

Рисунок 5: Настройка IP-адреса на PC2-user

3.6 Моделирование простейшей сети на базе коммутатора в GNS3

```
VPCS> ping 192.168.1.11/24

host (192.168.1.11) not reachable

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.1.12/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

VPCS> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.384 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.506 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.507 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.568 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.644 ms
```

Рисунок 6: Проверка связи между PC1 и PC2

3.7 Моделирование простейшей сети на базе коммутатора в GNS3

```
VPCS> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=0.516 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=0.483 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=0.506 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=7.082 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=0.673 ms
```

Рисунок 7: Успешный ping между узлами

3.8 Моделирование простейшей сети на базе коммутатора в GNS3

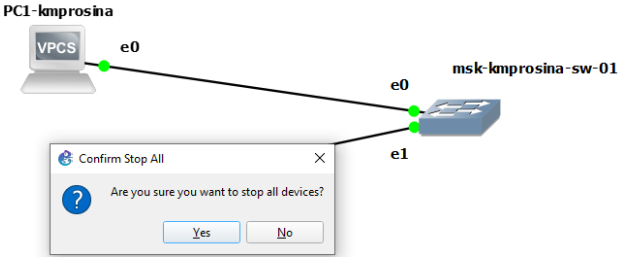
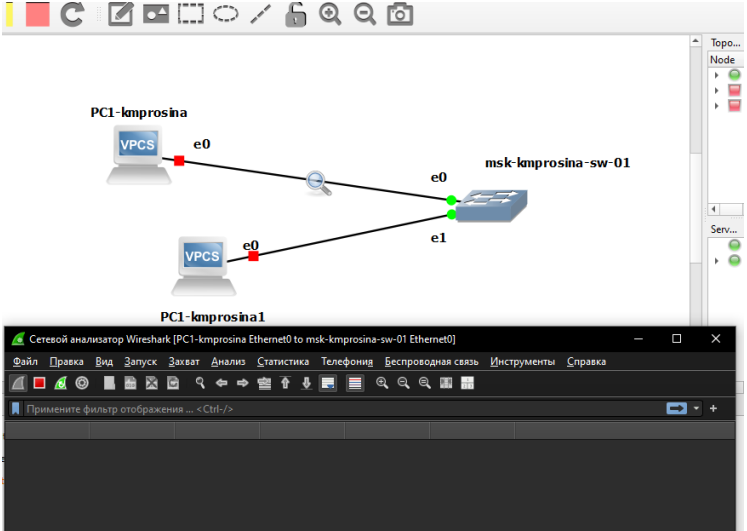


Рисунок 8: Остановка всех узлов проекта

3.9 Анализ трафика в GNS3 посредством Wireshark



3.10 Анализ трафика в GNS3 посредством Wireshark

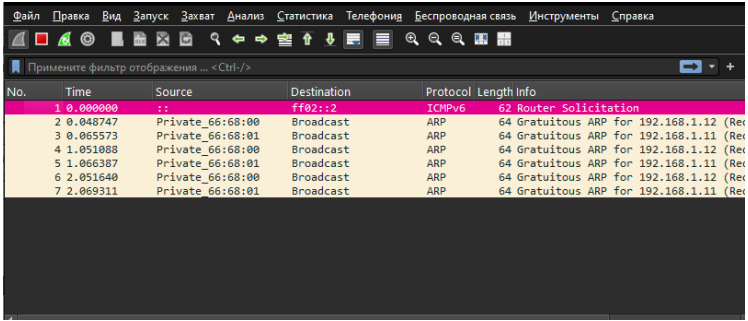


Рисунок 10: ARP-запросы в Wireshark

3.11 Анализ трафика в GNS3 посредством Wireshark

```
ping HOST [OPTION ...]
  Ping the network HOST. HOST can be an ip address or name
    Options:
     -1             ICMP mode, default
     -2             UDP mode
     -3             TCP mode
     -c count       Packet count, default 5
     -D             Set the Don't Fragment bit
     -f FLAG        Tcp header FLAG |C|E|U|A|P|R|S|F|
                                bits |7 6 5 4 3 2 1 0|
     -i ms          Wait ms milliseconds between sending each packet
     -l size        Data size
     -P protocol    Use IP protocol in ping packets
                      1 - ICMP (default), 17 - UDP, 6 - TCP
     -p port        Destination port
     -s port        Source port
     -T ttl         Set ttl, default 64
     -t             Send packets until interrupted by Ctrl+C
     -w ms          Wait ms milliseconds to receive the response

  Notes: 1. Using names requires DNS to be set.
         2. Use Ctrl+C to stop the command.

VPCS>
```

Рисунок 11: Подготовка к отправке ICMP-запроса

3.12 Анализ трафика в GNS3 посредством Wireshark

```
VPCS> ping 192.168.1.11 -l

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=0.995 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.624 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=0.547 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.788 ms
```

Рисунок 12: Отправка ping на адрес 192.168.1.11

3.13 Анализ трафика в GNS3 посредством Wireshark

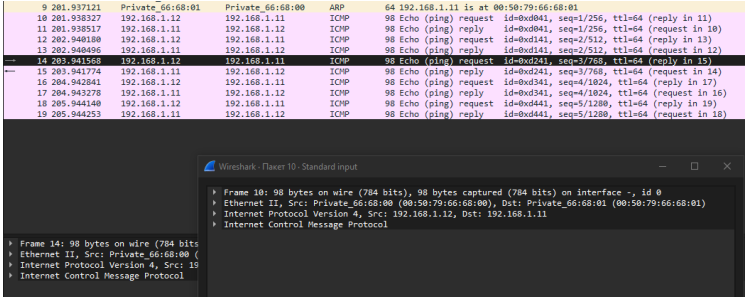


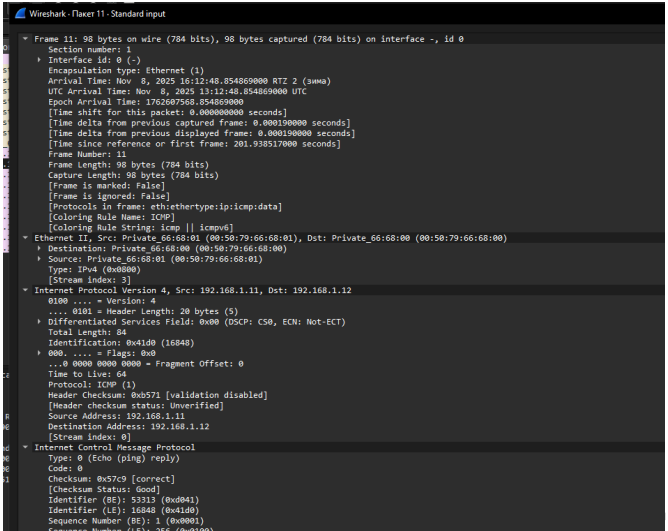
Рисунок 13: ICMP-запрос в Wireshark

3.14 Анализ трафика в GNS3 посредством Wireshark

Wireshark · Пакет 10 · Standard input

```
▾ Frame 10: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
    Section number: 1
  ▸ Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  8, 2025 16:12:48.854679000 RTZ 2 (зима)
    UTC Arrival Time: Nov  8, 2025 13:12:48.854679000 UTC
    Epoch Arrival Time: 1762607568.854679000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.001206000 seconds]
    [Time delta from previous displayed frame: 0.001206000 seconds]
    [Time since reference or first frame: 201.938327000 seconds]
    Frame Number: 10
    Frame Length: 98 bytes (784 bits)
    Capture Length: 98 bytes (784 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:icmp:data]
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
▾ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
  ▸ Destination: Private_66:68:01 (00:50:79:66:68:01)
  ▸ Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: IPv4 (0x0800)
    [Stream index: 3]
▾ Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x41d0 (16848)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0xb571 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.12
    Destination Address: 192.168.1.11
    [Stream index: 0]
▾ Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
```

3.15 Анализ трафика в GNS3 посредством Wireshark



3.16 Анализ трафика в GNS3 посредством Wireshark

```
VPCS> ping 192.168.1.11 -2

84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=0.877 ms
84 bytes from 192.168.1.11 udp_seq=2 ttl=64 time=0.340 ms
84 bytes from 192.168.1.11 udp_seq=3 ttl=64 time=0.504 ms
84 bytes from 192.168.1.11 udp_seq=4 ttl=64 time=0.590 ms
84 bytes from 192.168.1.11 udp_seq=5 ttl=64 time=0.535 ms
^Z
VPCS>
```

Рисунок 16: Просмотр опций команды ping

3.17 Анализ трафика в GNS3 посредством Wireshark

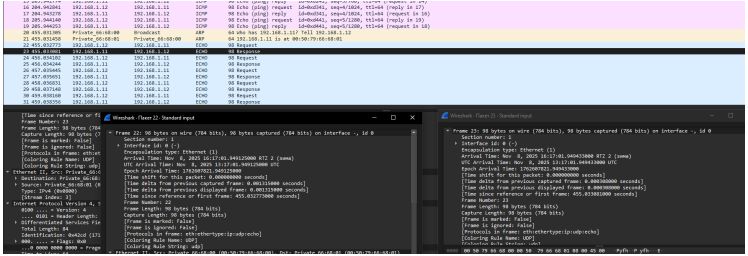


Рисунок 17: Отправка UDP-эхо-запроса

3.18 Анализ трафика в GNS3 посредством Wireshark

```
Frame 22: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
    Section number: 1
  ▸ Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  8, 2025 16:17:01.949125000 RTZ 2 (зима)
    UTC Arrival Time: Nov  8, 2025 13:17:01.949125000 UTC
    Epoch Arrival Time: 1762607821.949125000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.001315000 seconds]
    [Time delta from previous displayed frame: 0.001315000 seconds]
    [Time since reference or first frame: 455.032773000 seconds]
    Frame Number: 22
    Frame Length: 98 bytes (784 bits)
    Capture Length: 98 bytes (784 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:udp:echo]
    [Coloring Rule Name: UDP]
    [Coloring Rule String: udp]
▾ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
  ▸ Destination: Private_66:68:01 (00:50:79:66:68:01)
  ▸ Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: IPv4 (0x0800)
    [Stream index: 3]
▾ Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x42cd (17101)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: UDP (17)
    Header Checksum: 0xb464 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.12
    Destination Address: 192.168.1.11
    [Stream index: 0]
▾ User Datagram Protocol, Src Port: 50742, Dst Port: 7
    Source Port: 50742
    Destination Port: 7
    Length: 64
```

3.19 Анализ трафика в GNS3 посредством Wireshark

Wireshark · Пакет 23 · Standard input

```
▼ Frame 23: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
    Section number: 1
  ▶ Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  8, 2025 16:17:01.949433000 RTZ 2 (зима)
    UTC Arrival Time: Nov  8, 2025 13:17:01.949433000 UTC
    Epoch Arrival Time: 1762607821.949433000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000308000 seconds]
    [Time delta from previous displayed frame: 0.000308000 seconds]
    [Time since reference or first frame: 455.033081000 seconds]
    Frame Number: 23
    Frame Length: 98 bytes (784 bits)
    Capture Length: 98 bytes (784 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:udp:echo]
    [Coloring Rule Name: UDP]
    [Coloring Rule String: udp]
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
  ▶ Destination: Private_66:68:00 (00:50:79:66:68:00)
  ▶ Source: Private_66:68:01 (00:50:79:66:68:01)
    Type: IPv4 (0x0800)
    [Stream index: 3]
▼ Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.12
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x42cd (17101)
  ▶ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: UDP (17)
    Header Checksum: 0xb464 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.11
    Destination Address: 192.168.1.12
    [Stream index: 0]
▼ User Datagram Protocol, Src Port: 7, Dst Port: 50742
    Source Port: 7
    Destination Port: 50742
    Length: 64
    Checksum: 0x1a3f [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
```

3.20 Анализ трафика в GNS3 посредством Wireshark

```
VPCS> ping 192.168.1.11 -3

Connect   7@192.168.1.11 seq=1 ttl=64 time=1.110 ms
SendData  7@192.168.1.11 seq=1 ttl=64 time=1.202 ms
Close     7@192.168.1.11 seq=1 ttl=64 time=2.120 ms
Connect   7@192.168.1.11 seq=2 ttl=64 time=1.115 ms
SendData  7@192.168.1.11 seq=2 ttl=64 time=1.054 ms
Close     7@192.168.1.11 seq=2 ttl=64 time=2.532 ms
Connect   7@192.168.1.11 seq=3 ttl=64 time=1.117 ms
SendData  7@192.168.1.11 seq=3 ttl=64 time=1.173 ms
Close     7@192.168.1.11 seq=3 ttl=64 time=2.367 ms
Connect   7@192.168.1.11 seq=4 ttl=64 time=3.567 ms
SendData  7@192.168.1.11 seq=4 ttl=64 time=1.097 ms
Close     7@192.168.1.11 seq=4 ttl=64 time=4.184 ms
^ZConnect   7@192.168.1.11 seq=5 ttl=64 time=1.135 ms
SendData  7@192.168.1.11 seq=5 ttl=64 time=1.148 ms
Close     7@192.168.1.11 seq=5 ttl=64 time=2.206 ms
```

Рисунок 20: Просмотр опций TCP ping

3.21 Анализ трафика в GNS3 посредством Wireshark

33	789.007742	Private_66:68:01	Private_66:68:00	ARP	64 192.168.1.11 is at 00:50:79:66:68:01
34	789.008468	192.168.1.12	192.168.1.11	TCP	74 27141 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1762608155 TSecr=0 WS=2
35	789.008750	192.168.1.11	192.168.1.12	TCP	54 7 → 27141 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
36	789.009553	192.168.1.12	192.168.1.11	TCP	66 27141 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1762608155 TSecr=0
37	789.009785	192.168.1.12	192.168.1.11	ECHO	122 Request
38	789.010063	192.168.1.11	192.168.1.12	TCP	54 7 → 27141 [ACK] Seq=1 Ack=57 Win=2920 Len=0
39	789.012549	192.168.1.11	192.168.1.12	TCP	66 27141 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1762608155 TSecr=0
40	789.012894	192.168.1.11	192.168.1.12	TCP	54 7 → 27141 [ACK] Seq=1 Ack=58 Win=2920 Len=0
41	789.012933	192.168.1.11	192.168.1.12	TCP	54 7 → 27141 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
42	789.015051	192.168.1.12	192.168.1.11	TCP	66 27141 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1762608155 TSecr=0
43	790.015250	192.168.1.12	192.168.1.11	TCP	74 [TCP Port numbers reused] 27141 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1762608156 TSecr=0 WS=2
44	790.015471	192.168.1.11	192.168.1.12	TCP	54 7 → 27141 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
45	790.016561	192.168.1.12	192.168.1.11	TCP	66 27141 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1762608156 TSecr=0

Рисунок 21: Отправка TCP-эхо-запроса

3.22 Анализ трафика в GNS3 посредством Wireshark

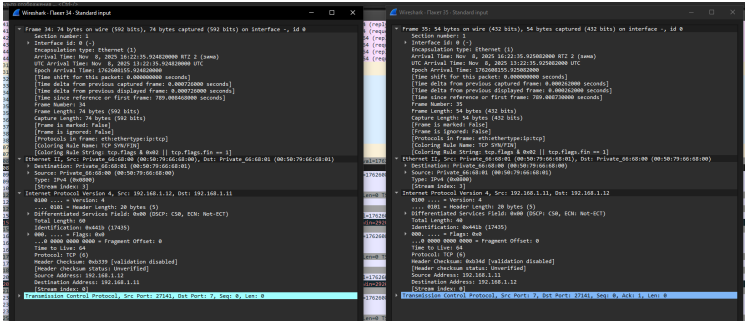


Рисунок 22: TCP SYN-пакет в Wireshark

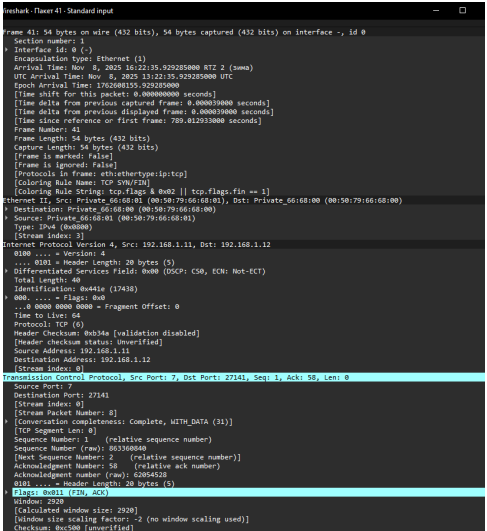
3.23 Анализ трафика в GNS3 посредством Wireshark

```
▼ Frame 36: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on inter
    Section number: 1
  ▶ Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  8, 2025 16:22:35.925905000 RTZ 2 (зима)
    UTC Arrival Time: Nov  8, 2025 13:22:35.925905000 UTC
    Epoch Arrival Time: 1762608155.925905000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000023000 seconds]
    [Time delta from previous displayed frame: 0.000023000 seconds]
    [Time since reference or first frame: 789.009553000 seconds]
    Frame Number: 36
    Frame Length: 66 bytes (528 bits)
    Capture Length: 66 bytes (528 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:0
  ▶ Destination: Private_66:68:01 (00:50:79:66:68:01)
  ▶ Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: IPv4 (0x0800)
    [Stream index: 3]
▼ Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 52
    Identification: 0x441c (17436)
  ▶ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0xb340 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.12
    Destination Address: 192.168.1.11
    [Stream index: 0]
▼ Transmission Control Protocol, Src Port: 27141, Dst Port: 7, Seq: 1, Ack: 1,
    Source Port: 27141
    Destination Port: 7
    [Stream index: 0]
    [Stream Packet Number: 3]
  ▶ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 62054471
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 863360840
    1000 .... = Header Length: 32 bytes (8)
```

3.24 Анализ трафика в GNS3 посредством Wireshark

```
Frame 40: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface -, id 0
    Section number: 1
  ▸ Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov  8, 2025 16:22:35.929246000 RTZ 2 (зима)
    UTC Arrival Time: Nov  8, 2025 13:22:35.929246000 UTC
    Epoch Arrival Time: 1762608155.929246000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000345000 seconds]
    [Time delta from previous displayed frame: 0.000345000 seconds]
    [Time since reference or first frame: 789.012894000 seconds]
    Frame Number: 40
    Frame Length: 54 bytes (432 bits)
    Capture Length: 54 bytes (432 bits)
    [frame is marked: false]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp]
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
  ▸ Destination: Private_66:68:00 (00:50:79:66:68:00)
  ▸ Source: Private_66:68:01 (00:50:79:66:68:01)
    Type: IPv4 (0x0800)
    [Stream index: 3]
Internet Protocol Version 4, Src: 192.168.1.11, Dst: 192.168.1.12
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 40
    Identification: 0x441e (17438)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0xb34a [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.11
    Destination Address: 192.168.1.12
    [Stream index: 0]
Transmission Control Protocol, Src Port: 7, Dst Port: 27141, Seq: 1, Ack: 58, Len: 0
    Source Port: 7
    Destination Port: 27141
    [Stream index: 0]
    [Stream Packet Number: 7]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 863360840
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 58    (relative ack number)
    Acknowledgment number (raw): 62054528
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x010 (ACK)
    Window: 2920
    [Calculated window size: 2920]
    [Window size scaling factor: -2 (no window scaling used)]
```

3.25 Анализ трафика в GNS3 посредством Wireshark



3.26 Анализ трафика в GNS3 посредством Wireshark

15	203.941774	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xd241, seq=3/768, ttl=64 (request in 14)
16	204.942841	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xd341, seq=4/1024, ttl=64 (reply in 17)
17	204.943278	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xd341, seq=4/1024, ttl=64 (request in 16)
18	205.944140	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xd441, seq=5/1280, ttl=64 (reply in 19)
19	205.944253	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xd441, seq=5/1280, ttl=64 (request in 18)
20	455.031305	Private_66:68:00	Broadcast	ARP	64	who has 192.168.1.11? Tell 192.168.1.12
21	455.031458	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.11 is at 00:50:79:66:68:01
22	455.032773	192.168.1.12	192.168.1.11	ECHO	98	Request
23	455.033081	192.168.1.11	192.168.1.12	ECHO	98	Response
24	456.034102	192.168.1.12	192.168.1.11	ECHO	98	Request
25	456.034244	192.168.1.11	192.168.1.12	ECHO	98	Response
26	457.035445	192.168.1.12	192.168.1.11	ECHO	98	Request
27	457.035651	192.168.1.11	192.168.1.12	ECHO	98	Response
28	458.036831	192.168.1.12	192.168.1.11	ECHO	98	Request
29	458.037148	192.168.1.11	192.168.1.12	ECHO	98	Response
30	459.038160	192.168.1.12	192.168.1.11	ECHO	98	Request
31	459.038356	192.168.1.11	192.168.1.12	ECHO	98	Response
32	789.007436	Private_66:68:00	Broadcast	ARP	64	who has 192.168.1.11? Tell 192.168.1.12
33	789.007742	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.11 is at 00:50:79:66:68:01
34	789.008468	192.168.1.12	192.168.1.11	TCP	74	27141 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1762608155 TSecr=0 WS=2
35	789.008730	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
36	789.009553	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1762608155 TSecr=0
37	789.009785	192.168.1.12	192.168.1.11	ECHO	122	Request
38	789.010063	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=57 Win=2920 Len=0
39	789.012549	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1762608155 TSecr=0
40	789.012894	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=58 Win=2920 Len=0
41	789.012933	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
42	789.015051	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1762608155 TSecr=0
43	790.015250	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 27141 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1762608156 TSe
44	790.015471	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
45	790.016361	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1762608156 TSecr=0
46	790.016665	192.168.1.12	192.168.1.11	ECHO	122	Request
47	790.016820	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=57 Win=2920 Len=0
48	790.017803	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1762608156 TSecr=0
49	790.017978	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=58 Win=2920 Len=0
50	790.018008	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
51	790.020492	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1762608156 TSecr=0
52	791.020755	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 27141 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1762608157 TSe
53	791.021019	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
54	791.023081	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1762608157 TSecr=0
55	791.023692	192.168.1.12	192.168.1.11	ECHO	122	Request
56	791.023857	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=57 Win=2920 Len=0
57	791.025239	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1762608157 TSecr=0
58	791.025481	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=58 Win=2920 Len=0
59	791.025514	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
60	791.027808	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1762608157 TSecr=0
61	792.028202	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 27141 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1762608158 TSe
62	792.030644	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
63	792.031916	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1762608158 TSecr=0
64	792.034924	192.168.1.12	192.168.1.11	ECHO	122	Request
65	792.035252	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=57 Win=2920 Len=0
66	792.038947	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1762608158 TSecr=0
67	792.040274	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [ACK] Seq=1 Ack=58 Win=2920 Len=0
68	792.040392	192.168.1.11	192.168.1.12	TCP	54	7 → 27141 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
69	792.043415	192.168.1.12	192.168.1.11	TCP	66	27141 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1762608158 TSecr=0

3.27 Моделирование простейшей сети на базе маршрутизатора FRR

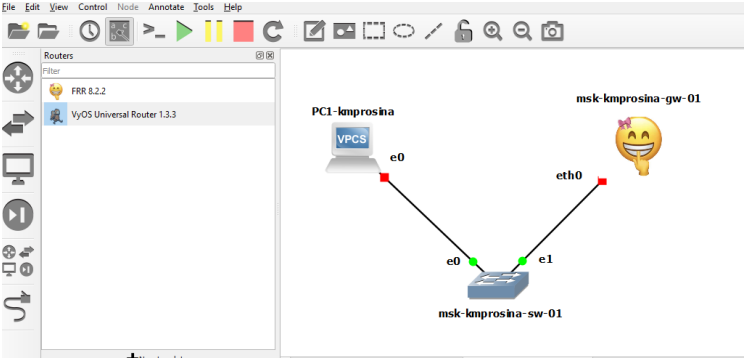
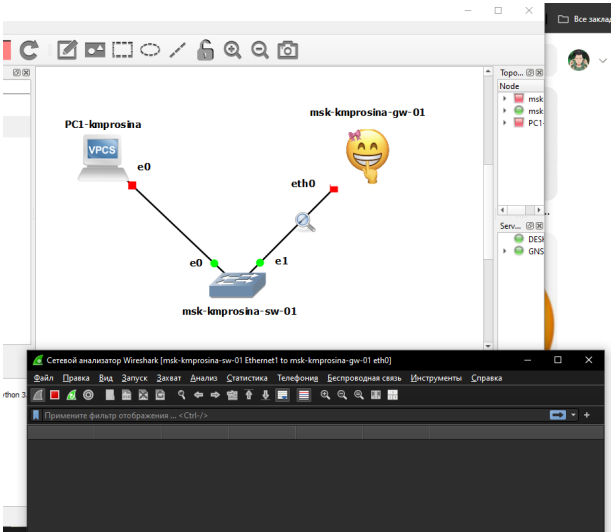


Рисунок 27: Топология сети с маршрутизатором FRR

3.28 Моделирование простейшей сети на базе маршрутизатора FRR



3.29 Моделирование простейшей сети на базе маршрутизатора FRR

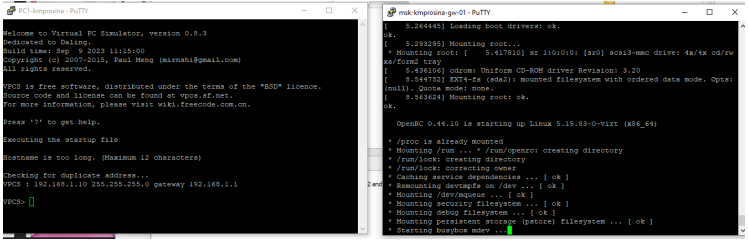


Рисунок 29: Консоли всех устройств проекта

3.30 Моделирование простейшей сети на базе маршрутизатора FRR

PC1-mvchetvergova - PuTTY

```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500
```

3.31 Моделирование простейшей сети на базе маршрутизатора FRR

```
frr#
frr# configure terminal
frr(config)# hostname msk-kmprosina-gw-01
msk-kmprosina-gw-01(config)# exit
msk-kmprosina-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-kmprosina-gw-01#
msk-kmprosina-gw-01# configure terminal
msk-kmprosina-gw-01(config)# interface eth0
msk-kmprosina-gw-01(config-if)# ip address 192.168.1.1/24
msk-kmprosina-gw-01(config-if)# no shutdown
msk-kmprosina-gw-01(config-if)# exit
msk-kmprosina-gw-01(config)# write memory
% Unknown command: write memory
msk-kmprosina-gw-01(config)#
```

Рисунок 31: Настройка интерфейса маршрутизатора FRR

3.32 Моделирование простейшей сети на базе маршрутизатора FRR

```
msk-kmprosina-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-kmprosina-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-kmprosina-gw-01# show interface brief
Interface       Status  VRF             Addresses
---------       ------  ---             ---------
eth0            up      default         192.168.1.1/24
eth1            down    default
eth2            down    default
eth3            down    default
eth4            down    default
eth5            down    default
eth6            down    default
eth7            down    default
lo              up      default
```

3.33 Моделирование простейшей сети на базе маршрутизатора FRR

```
VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=5.054 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.661 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.648 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=3.120 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.621 ms
```

Рисунок 33: Проверка связи с маршрутизатором

3.34 Моделирование простейшей сети на базе маршрутизатора FRR

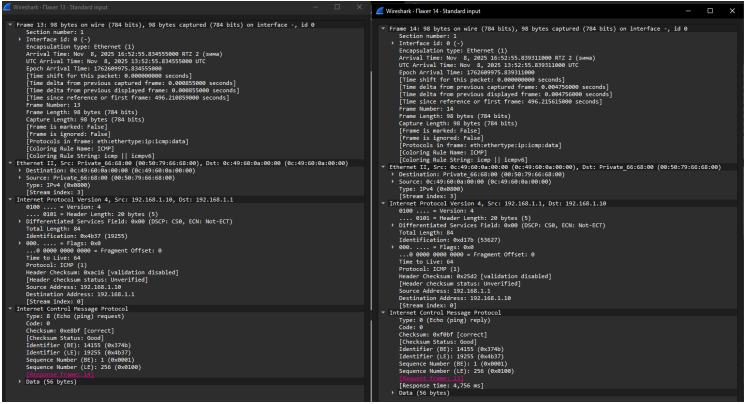


Рисунок 34: Анализ трафика в Wireshark

3.35 Моделирование простейшей сети на базе маршрутизатора FRR

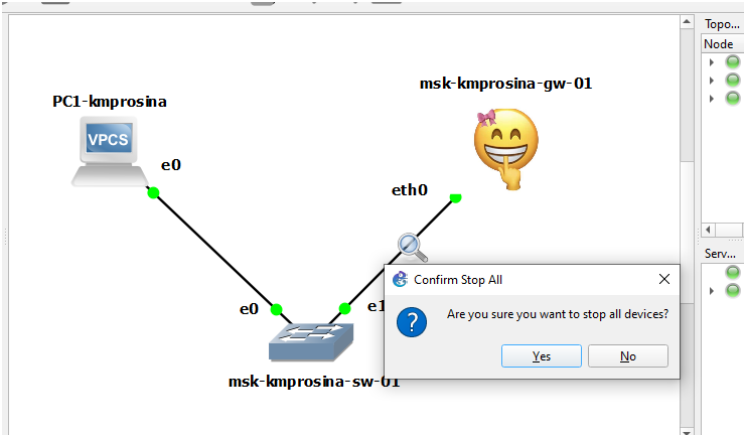


Рисунок 35: Остановка захвата трафика и устройств

3.36 Моделирование простейшей сети на базе маршрутизатора VyOS

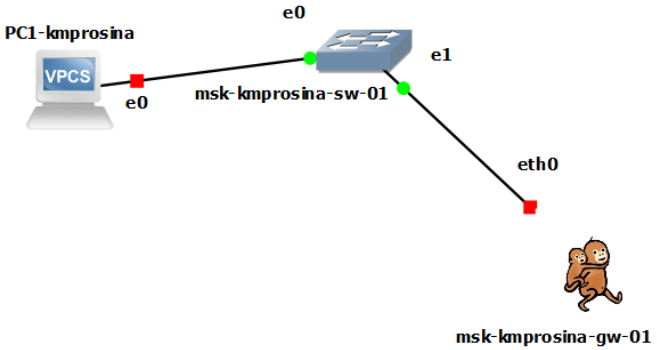
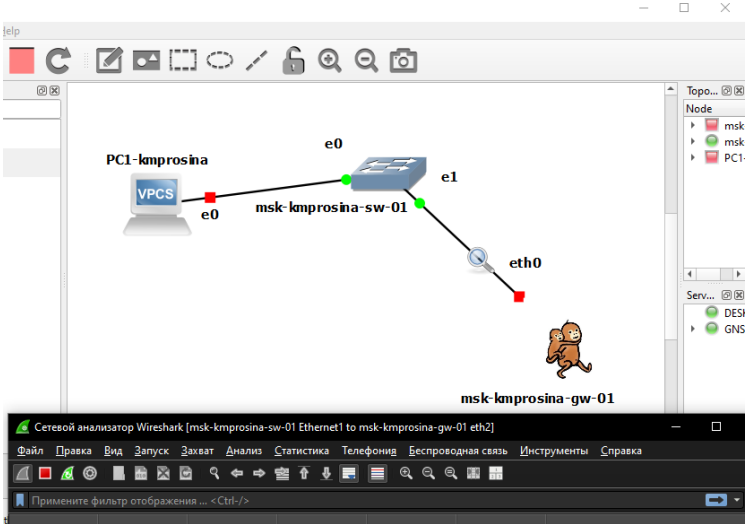


Рисунок 36: Топология сети с маршрутизатором VyOS

3.37 Моделирование простейшей сети на базе маршрутизатора VyOS



3.38 Моделирование простейшей сети на базе маршрутизатора VyOS

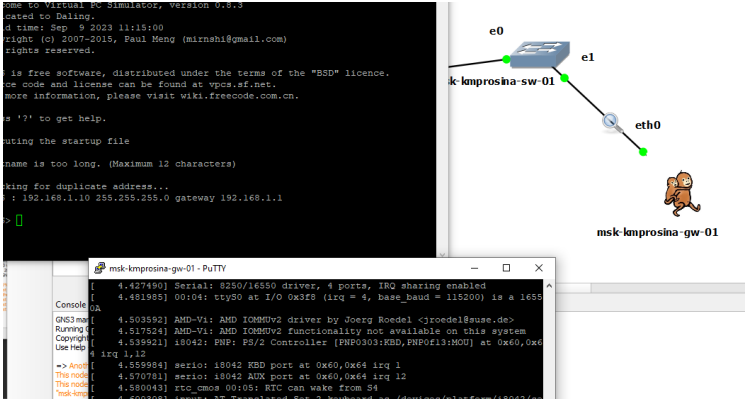


Рисунок 38: Консоли устройств в проекте с VyOS

3.39 Моделирование простейшей сети на базе маршрутизатора VyOS

```
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS>
VPCS>
VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

VPCS>
```

3.40 Моделирование простейшей сети на базе маршрутизатора VyOS

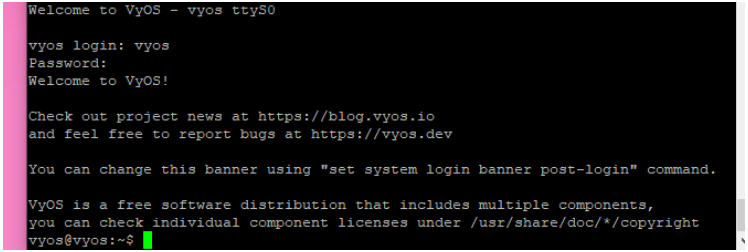


Рисунок 40: Вход в систему VyOS

3.41 Моделирование простейшей сети на базе маршрутизатора VyOS

```
vyos@vyos:~$
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$
```

Рисунок 41: Переход в режим конфигурации

3.42 Моделирование простейшей сети на базе маршрутизатора VyOS

```
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos#
```

Рисунок 42: Установка hostname маршрутизатора

3.43 Моделирование простейшей сети на базе маршрутизатора VyOS

```
VyOS is a free software distribution that includes multiple componen
you can check individual component licenses under /usr/share/doc/*/c
vyos@vyos:~$
vyos@vyos:~$
vyos@vyos:~$
vyos@vyos:~$ set system host-name msk-kmprosina-gw-01

  Invalid command: set system [host-name]

vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ set system host-name msk-kmprosina-gw-01

  Invalid command: set system [host-name]

vyos@vyos:~$ configure
[edit]
vyos@vyos#
[edit]
vyos@vyos# set system host-name msk-kmprosina-gw-01
[edit]
vyos@vyos#
```

3.44 Моделирование простейшей сети на базе маршрутизатора VyOS

```
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos#
```

Рисунок 44: Просмотр внесенных изменений

3.45 Моделирование простейшей сети на базе маршрутизатора VyOS

```
vyos@vyos:~$ configure
[edit]
vyos@vyos#
[edit]
vyos@vyos# set system host-name msk-kmprosina-gw-01
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-kmprosina-gw-01
```

Рисунок 45: Применение изменений конфигурации

3.46 Моделирование простейшей сети на базе маршрутизатора VyOS

```
[edit]
vyos@vyos# commit
[ interfaces ethernet eth0 ]
Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address dhcp
+    address 192.168.1.1/24
     hw-id 0c:49:61:03:00:00
 }
 ethernet eth1 {
     hw-id 0c:49:61:03:00:01
 }
 ethernet eth2 {
     hw-id 0c:49:61:03:00:02
 }
 ethernet eth3 {
     hw-id 0c:49:61:03:00:03
 }
 ethernet eth4 {
     hw-id 0c:49:61:03:00:04
 }
 ethernet eth5 {
     hw-id 0c:49:61:03:00:05
 }
 ethernet eth6 {
     hw-id 0c:49:61:03:00:06
 }
 ethernet eth7 {

[1]+  Stopped                 cli-shell-api "${opts[@]}" -- showCfg "${args[@]}"
 | eval "${VYATTA_PAGER:-cat}"
[edit]
vyos@vyos#
[edit]
vyos@vyos#
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24

  Configuration path: [interfaces ethernet eth0 address 192.168.1.1/24] already
```

3.47 Моделирование простейшей сети на базе маршрутизатора VyOS

```
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address 192.168.1.1/24
     hw-id 0c:49:61:03:00:00
 }
 ethernet eth1 {
     hw-id 0c:49:61:03:00:01
 }
 ethernet eth2 {
     hw-id 0c:49:61:03:00:02
 }
 ethernet eth3 {
     hw-id 0c:49:61:03:00:03
 }
 ethernet eth4 {
     hw-id 0c:49:61:03:00:04
 }
 ethernet eth5 {
     hw-id 0c:49:61:03:00:05
 }
 ethernet eth6 {
     hw-id 0c:49:61:03:00:06
 }
 ethernet eth7 {
     hw-id 0c:49:61:03:00:07

[2]+  Stopped                 cli-shell-api "${opts[@]}" -- showCfg "${args[@]}" | eval "${VYATTA_PAGER
:-cat}"
[edit]
vyos@vyos# exit
Warning: configuration changes have not been saved.
exit
There are stopped jobs.
DEBUG vexit_internal: calling getEditResetEnv() without config session
[edit]
vyos@vyos#
```

3.48 Моделирование простейшей сети на базе маршрутизатора VyOS

```
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All rights reserved.

VPCS is free software, distributed under the terms
Source code and license can be found at vpcs.sf.net
For more information, please visit wiki.freecode.co

Press '?' to get help.

Executing the startup file


Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1

VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=3.
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.
^Z84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=
```
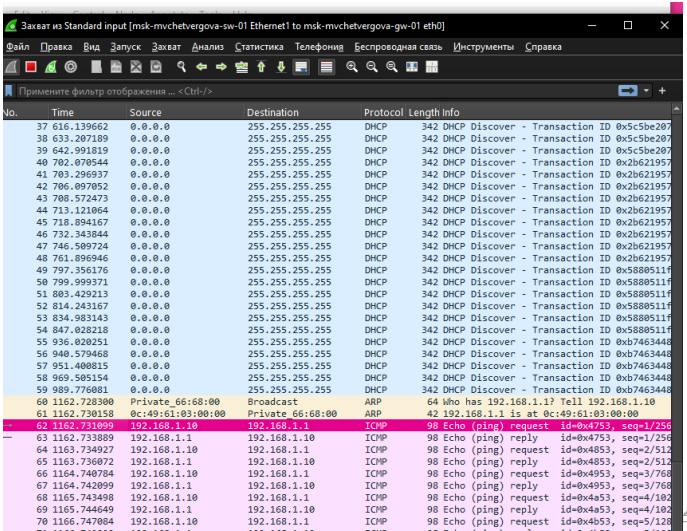

3.49 Моделирование простейшей сети на базе маршрутизатора VyOS



3.50 Моделирование простейшей сети на базе маршрутизатора VyOS

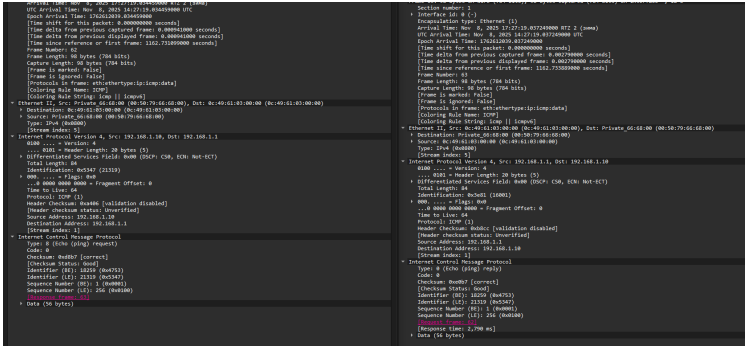


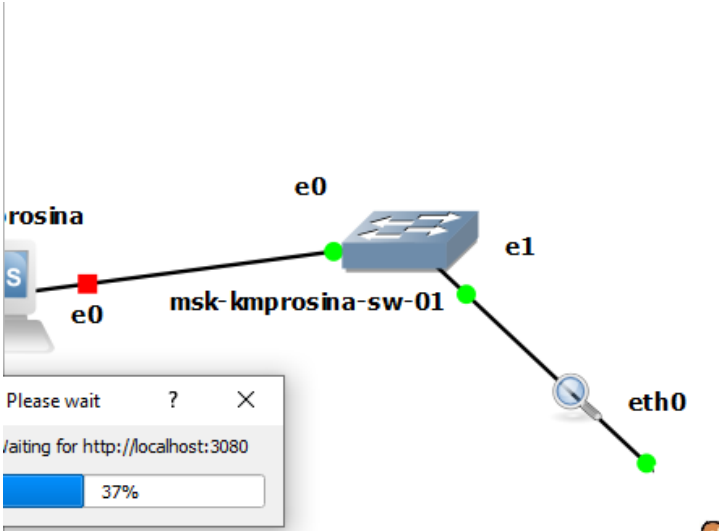
Рисунок 50: Трафик ICMP в сети с VyOS

3.51 Моделирование простейшей сети на базе маршрутизатора VyOS

60	1162.728300	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.
61	1162.730158	0c:49:61:03:00:00	Private_66:68:00	ARP	42	192.168.1.1 is at 0c:49:61:03:00:00
62	1162.731099	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x4753, seq=
63	1162.733889	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x4753, seq=
64	1163.734927	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x4853, seq=
65	1163.736072	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x4853, seq=
66	1164.740784	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x4953, seq=
67	1164.742099	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x4953, seq=
68	1165.743498	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x4a53, seq=
69	1165.744649	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x4a53, seq=
70	1166.747084	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x4b53, seq=
71	1166.749220	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x4b53, seq=
72	1167.742519	0c:49:61:03:00:00	Private_66:68:00	ARP	42	Who has 192.168.1.10? Tell 192.168.1
73	1167.743270	Private_66:68:00	0c:49:61:03:00:00	ARP	42	192.168.1.10 is at 00:50:79:66:68:00

Рисунок 51: Остановка захвата трафика

3.52 Моделирование простейшей сети на базе маршрутизатора VyOS



3.53 Моделирование простейшей сети на базе маршрутизатора VyOS

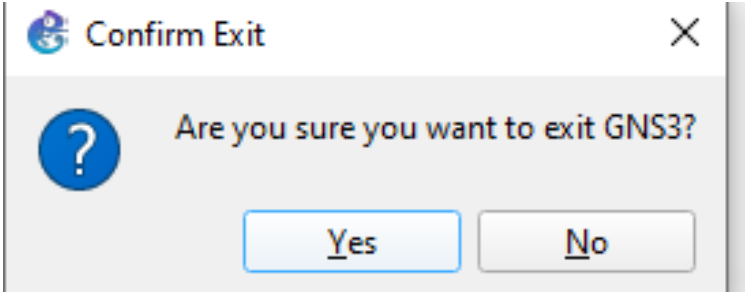


Рисунок 53: Завершение работы с GNS3

Раздел 4

4. Выводы

4.1 Выводы

В ходе выполнения лабораторной работы №5 я построила простейшие моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.